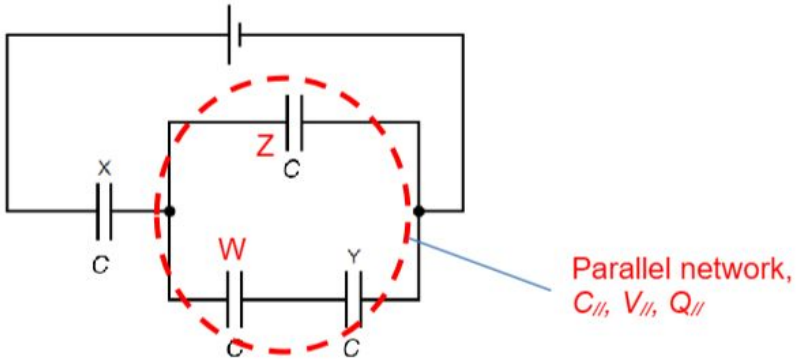
Hence, the maximum time t = (12)(60)/(36) = 20 h

23 A



For ease of reference, let's label all the capacitors as above.

The equivalent capacitance of the parallel network of capacitors on the right:

$$C_{//} = C + (\frac{1}{C} + \frac{1}{C})^{-1} = \frac{3}{2}C$$

Capacitor X is in series with the whole right-hand network, so $Q_{//} = Q_X = CV$.

Hence, $V_{//} = \frac{Q_{//}}{(\frac{3}{2}C)} = \frac{Q_X}{(\frac{3}{2}C)} = \frac{CV}{(\frac{3}{2}C)} = \frac{2}{3}V$

In the lower branch of the parallel circuit, C and C are in series which means they carry the same charge. Same charge and same capacitance → same potential difference.

$$V_Y = = \frac{V}{3}$$